

Joshua Plummer | Curriculum Vitae

Montgomery, Alabama
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Professional Profile

Teaching-focused mathematics educator and curriculum developer expanding expertise into instructional technology, professional development, and modern learning systems. Grounded in adult learning theory and evidence-based course design, with a strong record of creating engaging, student-centered learning experiences. Recognized through repeated student-nominated teaching honors.

Education

Auburn University at Montgomery

Ed.D. Instructional Systems and Learning Sciences
Concentration: Instructional Technology

Exp. May 2028

University of South Alabama

M.S. Mathematics
Thesis: On Matroid Representability
B.S. Mathematics & Statistics
B.S. Physics

Aug. 2016 – May 2018

Aug. 2010 – May 2016

Aug. 2010 – May 2016

Experience

Auburn University at Montgomery

Lecturer

Aug. 2024 – Present

- Design and teach engaging, equitable, and outcomes-aligned curricula for gateway mathematics and statistics courses by utilizing collaborative and experiential approaches. Face-to-face and online modalities.
- Maintain student evaluation ratings of 4.8/5.0 or higher — among the highest in the Mathematics Department.
- Collaborate with committee peers to coordinate syllabi, assessments, and corequisite support for multi-section gateway courses.
- Review course-performance data, including student evaluations, DFW patterns, and self-produced surveys to refine instruction and student support.

Purdue University Global

Adjunct Instructor

June 2025 – Present

- Teach online undergraduate mathematics courses for a diverse, nontraditional student population.
- Facilitate learning through structured weekly modules, discussion-based engagement, timely individualized feedback, and virtual office hours.
- Use LMS analytics and targeted outreach to identify barriers to progress and support student persistence.

University of South Alabama

Adjunct Instructor

Aug. 2018 – Present

- Teach face-to-face, online, and hybrid undergraduate mathematics courses for a diverse student population.
- Provide clear, actionable feedback to support persistence and mastery in virtual learning environments.
- Use LMS analytics to identify at-risk students and deliver targeted, personalized intervention.

Graduate Teaching Assistant

Aug. 2016 – May 2018

- Led recitation and problem-solving sessions.
- Developed and graded weekly quizzes.
- Held weekly office hours and exam review sessions.

Lead Peer Academic Leader (PAL)

Aug. 2012 – May 2016

- Mentored students and connected them with academic-support resources.
- Supervised and evaluated a team of 5–6 PALs.
- Supported onboarding, performance reviews, interviewing, and selection of PALs.

St. Luke's Episcopal School

Upper School Teacher

Aug. 2019 – Apr. 2023

- Taught advanced and AP-level mathematics and science courses; established AP Calculus AB into the upper-school curriculum.
- Spearheaded the development of inquiry-based and active-learning curricula across departments. In particular, adopted the experience first, formalize later model to promote mathematical intuition.
- Sponsored the Mu Alpha Theta chapter and coached the school's eSports team.

Faulkner University

Adjunct Instructor

Jan. 2021 – May 2022

- Taught undergraduate mathematics courses in face-to-face format.
- Develop department curricula and adopt textbooks.
- Assessed student work and delivered timely feedback to support skill development and course success.

Selected Work in Instructional Design, Curriculum, and Faculty Development

AI-Enhanced Instructional Design

Auburn University at Montgomery

2026

Post-Test Error Diagnosis: A Hybrid AI-Human Instructional Model for Mathematics

- Designed a six-step post-test routine in which students choose an AI coach or a short meeting with me. The coach is instructed never to grade, give answers, or solve the problem; I assess the student's own Error Analysis.
- Tested prompt card v2.0 across eight rounds and v2.1 in twelve simulated equity-audit sessions. Current card v2.2 incorporates the audit's recommendations and has not yet been tested.
- Proposed a two-semester comparison study of diagnostic accuracy, near transfer, and retention. Student learning outcomes have not yet been measured.

Professional Learning Toolkit

Auburn University at Montgomery

2026

Accessible & Rigorous: Designing Mathematics Instruction for the Modern Learner

- Designed a professional-learning toolkit for mathematics faculty, grounded in adult learning theory, backward design, and accessibility with three job-embedded professional development sessions.
- Facilitated a one-on-one adaptation of the *Your Page, Rebuilt* professional development session, guiding a colleague through an accessible Canvas page redesign and used feedback to refine the session.

Co-requisite Workbook Development

Auburn University at Montgomery

2025

- Revised and redeveloped co-requisite workbooks for Elementary Statistics, Precalculus, and College Algebra.
- Reviewed content, refined explanations, and restructured activities for clarity and alignment with learning objectives.

Academic Materials Development

University of South Alabama

2015

- Spearheaded the project with the Office of Student Success to support students in historically challenging courses in mathematics, physics, and chemistry.
- Developed 100+ academic-support resources, including quick-reference guides and example videos.

Committees and Service

2026 – Present: Social Committee	<i>Chair</i>
2024 – Present: Course Coordination Committees	<i>Member</i>
2024 – Present: Co-requisite Support Committees	<i>Member</i>
2026 – Present: Online Course Integrity Committee	<i>Member</i>
2026 – Present: Hiring Committee	<i>Member</i>
2025 – Present: Alabama State-Wide High School Mathematics Contest	<i>Member</i>
2025 – Present: SK-Day Committee	<i>Member</i>
2025 – 2026: Social Committee	<i>Co-Chair</i>

Certifications and Professional Development

Google Data Analytics

Coursera

in progress

Google AI Fundamentals

Coursera

2026

Elements of AI: Introduction to AI and Building AI

ElementsofAI.com

2026

AI for Educators

Microsoft

2026

Generative AI for Educators with Gemini

Google

2026

NextGen Learning: AI Integration Series

Purdue University Global

2026

Certificate Series in Accessibility Practices for Microsoft Suite and Canvas

Auburn University at Montgomery

2026

Enhancing Online Education through Instructor-Created Video Content

Purdue University Global

2026

Flight School (Student Success & Peer Mentoring Training)

Auburn University at Montgomery

2026

Certificate in Applied Data Analytics

Auburn University at Montgomery

2025

HHMI Grant-Supported Teaching & Student Success Training

Auburn University at Montgomery

2024

Experience First, Formalize Later Workshop*Math Medic Foundation*

2022

AP Summer Institute: AP Calculus AB*Auburn University*

2020

Selected Awards and Honors

Student-Nominated Significant Impact Certificate of Appreciation*Auburn University at Montgomery*

2026

Auburn University at Montgomery

2025

Auburn University at Montgomery

2024

Conference of Southern Graduate Schools Outstanding Thesis Award*University of South Alabama*

2019

Sandra & Melton McLaurin Graduate Fellowship*University of South Alabama*

2017

Recognition of Dedication and Leadership as a Lead JagPAL*University of South Alabama*

2016

Chris Nash Mathematics Competition Scholarship*University of South Alabama*

2015

Induction into Sigma Pi Sigma*University of South Alabama*

2013

Research and Publications

Master's Thesis*University of South Alabama*

2017 – 2018

On Matroid Representability (advisor: Dr. Elena Pavelescu). Explored matroid representations, which culminated in a look at Rota's conjecture and original proofs of selected results.

Editorial on the Importance of General Education Requirements*University of South Alabama*

2016

Plummer, J. (2016). *The Importance of General Education*. *The Vanguard*, 59(9), 7. Written for a freshman seminar course.

Astronomy Research and Publication*University of South Alabama*

2014 – 2015

Studied visual binary stars via speckle interferometry under Dr. Kent Clark, part of a study led by Russell Genet. Genet, R., et al. (2015). *Kitt Peak Speckle Interferometry of Close Visual Binary Stars*. *Journal of Double Star Observations*, 11(1s).

Spectroscopy Research*University of South Alabama*

2011 – 2012

Used gamma-ray spectroscopy to date soil core samples via Cs-137, Pb-210, and other radioactive isotopes, under Dr. Justin Sanders.